

Abirham Getie

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EDUCATION

The George Washington University

Master of Science, Data Science | GPA: 3.93

Washington, DC

Expected May 2026

Relevant Coursework: Machine Learning, Data Mining, Big Data Analytics, Data Warehousing, Cloud Computing (AWS), NLP

Gujarat Technological University

Bachelor of Computer Engineering

Ahmedabad, India

June 2024

Teaching Assistant: Led lab sessions on data structures and algorithms for 60+ undergraduates.

EXPERIENCE

Ethio Telecom

Systems Analyst

Addis Ababa, Ethiopia

Jan 2022 – Jun 2024

- Optimized database indexing on 500K+ customer service logs and network records, cutting query response time by 20% and accelerating weekly IT operations reporting workflows.
- Built Python anomaly detection system that flagged IT failures from streaming network logs, replacing manual log review and shifting the team from reactive to proactive incident response.
- Built interactive Tableau dashboards visualizing network outage hotspots, giving the IT team a single source of truth for triaging field issues.

Gujarat Technological University

Research Assistant (Part-time)

Ahmedabad, India

Jan 2024 – May 2024

- Trained SVM and Random Forest classifiers for fake news detection, achieving 96.5% accuracy and improving over the prior baseline by 4 points; [published](#) in peer-reviewed engineering journal.
- Engineered text preprocessing and feature extraction pipeline (TF-IDF, n-grams) on multi-source news corpus; ran ablation studies to identify most predictive linguistic features.

Open Avenues Foundation — Build Fellowship

AI/ML Fellow (Part-time)

Remote (New York, NY)

Feb 2025 – May 2025

- Built AI-powered image retrieval system on Caltech-101 under industry mentorship of Kamalesh Kalirathinam, extracting 2048-dim embeddings from pretrained ResNet-50 and indexing with FAISS for fast nearest-neighbor search; [GitHub](#)
- Engineered OpenCV image quality pipeline detecting blur (Laplacian variance) and JPEG compression artifacts (entropy + block-boundary), filtering corrupted images before indexing.
- Validated embedding quality with t-SNE showing clear class separation; deployed Streamlit web app (Docker-containerized) for real-time visual search, presented to fellowship cohort.

PROJECTS

AI-Powered IT Ticket Triage & Knowledge Base Generation | GW IT Capstone

Sep 2025 – Apr 2026

Python, FastAPI, React, Next.js, TypeScript, PostgreSQL, pgvector, Azure OpenAI (GPT-4.1, ada-002), AWS EC2, Docker

- Built production-grade full-stack RAG application processing 62K+ real-world tickets, with FastAPI backend, React/Next.js frontend, and AWS EC2 deployment, automating knowledge work for an enterprise IT team.
- Designed agentic LLM system orchestrating GPT-4.1 across labeling, summarization, deduplication, and LLM-as-judge validation roles; generated 40 validated KB articles from 73 critical gaps with 3-stage quality gates.
- Built semantic search and clustering pipeline using ada-002 embeddings (1,536-dim) and pgvector for vector similarity search, with custom silhouette scoring identifying 175 distinct topic clusters.
- Shipped iteratively via Git/GitHub workflows with code review and CI checks; designed priority scoring (volume + recency + gap severity + evidence quality) so authors focus on highest-impact gaps first.

Financial News Sentiment Classification | [Demo](#)

Python, PyTorch, Hugging Face Transformers, Streamlit, Google Colab Pro (A100 GPU)

- Built end-to-end NLP pipeline classifying 1.8M financial headlines as positive/neutral/negative, validating FinBERT pseudo-labels against expert-annotated Financial PhraseBank with 97.2% agreement.
- Benchmarked 5 models (LogReg, BiLSTM, DistilBERT, BERT-Base, FinBERT) plus soft-voting ensemble; FinBERT achieved 95.7% test accuracy and 0.96 weighted F1, beating classical baseline by 15 points.
- Conducted qualitative analysis on contradictory headlines, exposing FinBERT's domain advantage on ambiguous phrasing; deployed Streamlit app for real-time multi-model inference and comparison.

Amharic Hate Speech Detection | [Published](#)

Python, NLP, Facepager API

- Developed ML classifier to detect Amharic hate speech on social media, trained on a self-collected dataset of 10K+ posts gathered via Facepager API; published as peer-reviewed Springer book chapter.
- Deployed model with Ethiopian government partner to flag harmful content on monitored platforms, contributing to a measurable reduction in detected hate speech volume.

TECHNICAL SKILLS

LLMs & GenAI: GPT-4, Claude, RAG, Agentic Systems, LLM-as-Judge, Prompt Engineering, Fine-tuning, Hugging Face, OpenAI API, Embeddings

ML & CV/NLP: PyTorch, scikit-learn, BERT, DistilBERT, FinBERT, ResNet-50, FAISS, OpenCV, t-SNE, K-Means, NLTK, SpaCy

Full Stack & Cloud: Python, TypeScript, SQL, FastAPI, React, Next.js, REST APIs, PostgreSQL, MongoDB, AWS, Docker, GitHub Actions